Netflix Project - Problem Statement

With the rapid growth of OTT and streaming platforms, users are presented with thousands of movies and TV shows, making it difficult to discover content that matches their interests. Recommendation systems play a crucial role in enhancing user experience by analyzing historical user interactions and suggesting personalized content.

In this project, you will build a Movie Recommendation System using the Netflix Prize dataset. The objective is to analyze user ratings, movie information, and genre data to understand user preferences and generate personalized movie recommendations. The recommendation engine should be implemented using the Singular Value Decomposition (SVD) algorithm, a collaborative filtering technique that predicts user preferences based on latent factors learned from historical ratings.

The Movie Ratings dataset requires preprocessing, as both User IDs and Movie IDs are stored in the same column. Entries ending with a colon (:) represent User IDs, while the subsequent numeric entries until the next user identifier represent the Movie IDs associated with that user. Learners are expected to correctly transform this raw data into a structured format before performing analysis and model development.

In addition to building the recommendation model, learners should perform Exploratory Data Analysis (EDA) to understand rating distributions, user behavior, movie popularity, and genre-wise trends. The final solution should recommend movies tailored to individual users and evaluate the recommendation model using appropriate performance metrics.

Data Set Links -
https://drive.google.com/file/d/1nUAwFYxnGiPhFOgP4tiXwOkaL8BgCMQG/view

https://drive.google.com/file/d/1_QXVewqmoXBF5tlr4iOPYVs_2XqrVwnF/view